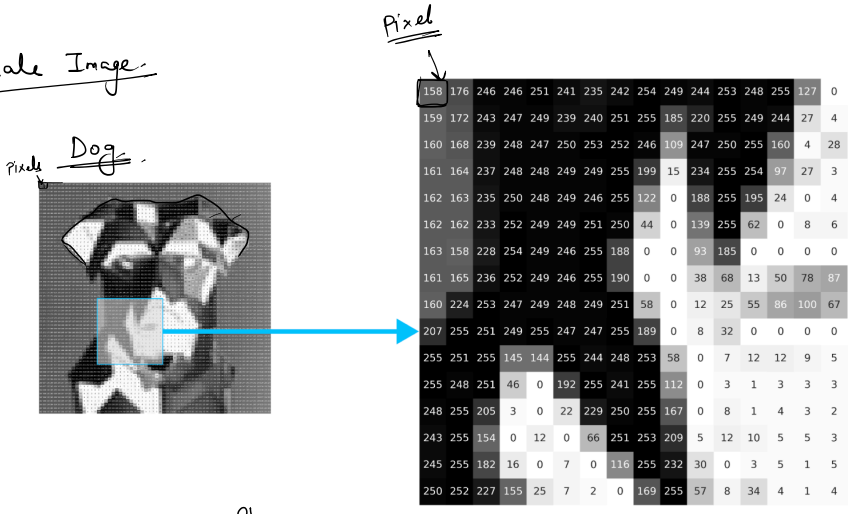
Computer Vision: Teaching Machines to See

Thursday, 29 October 2026 12:26 AM

* Computer Vision: It is a field of AI that enables machine to derive Meaningful information from Images.

Grayscale Image

Pixel



Dog.

255 → Dark. (B)

0 → Lighter (W)

B/W.

Pixel → (0-255).

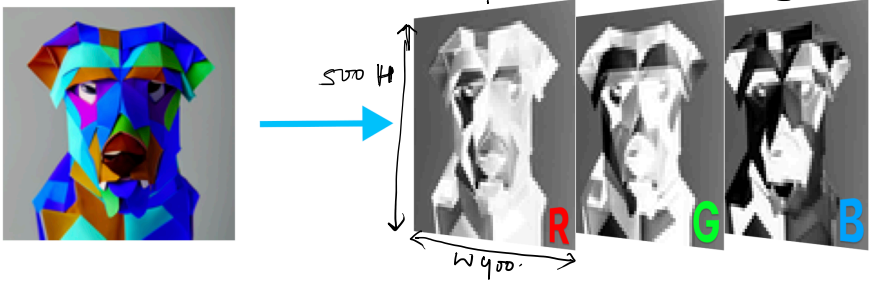
Color Image.

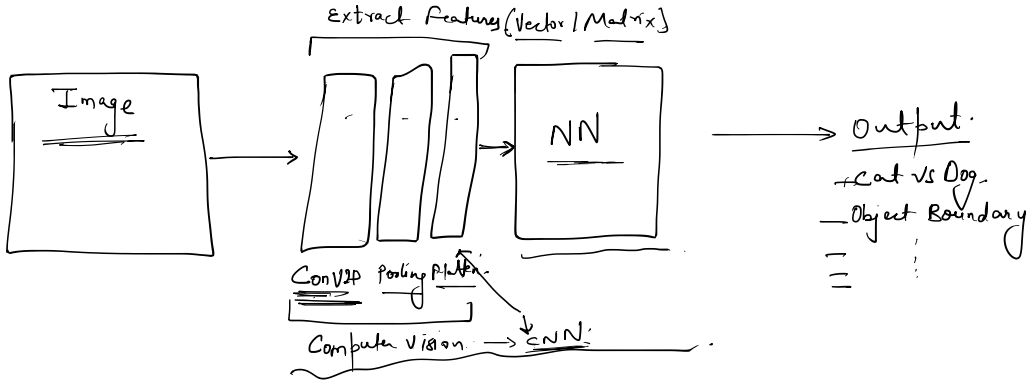
Red Blue

(R, G, B) → (C, H, W). → (3, 500, 400).

Green

3





Images → L1 → L2 → L3 → L4 → L5.

L1: High level feature extraction. (extracting edges)

L2: Creates shapes.

L3: Identify objects

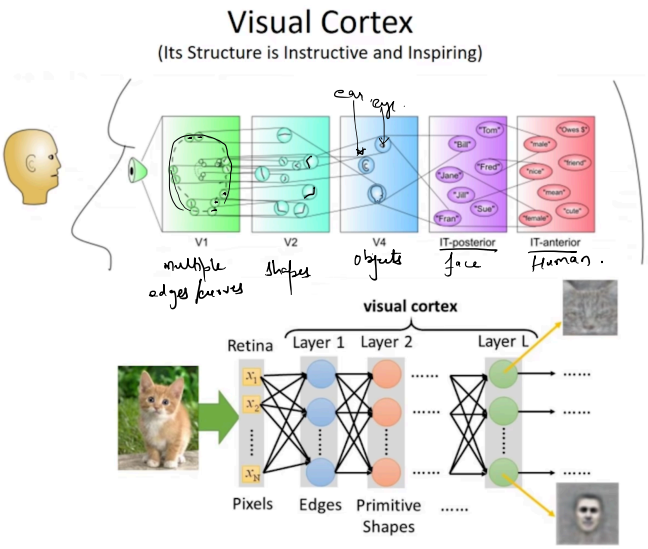
L4: Identify a scene

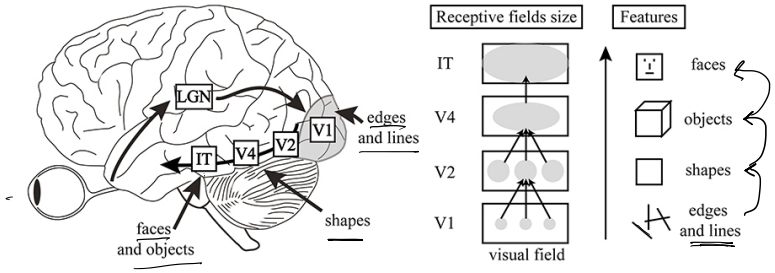
L5: Output.

lines / curves.

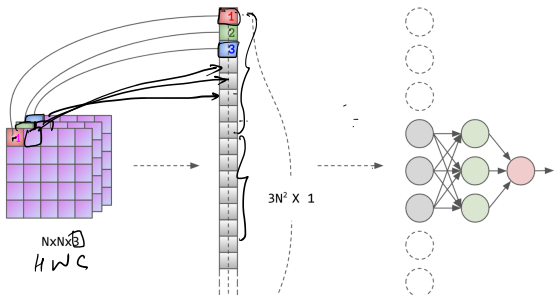
Use case : To identify content Quality score.

Restaurants → Food Image.





Q. Why can't we use ANN/ for Images ?
MLP





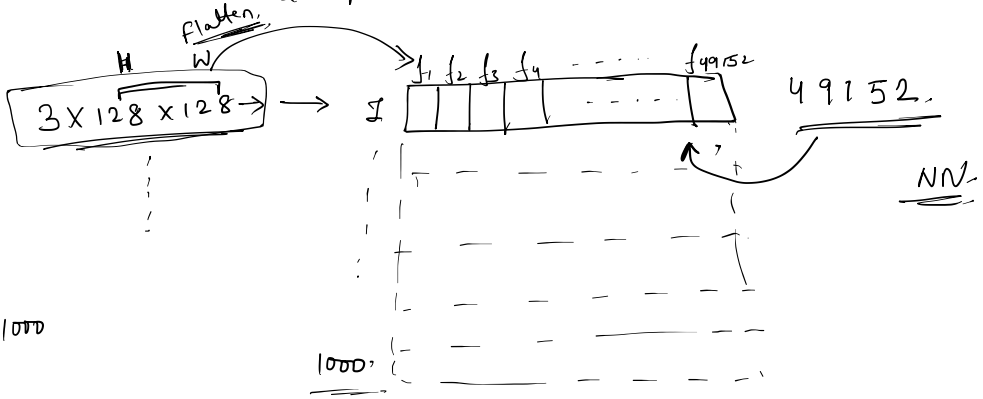
A NN.

1) Resizing : 400 x 233, 300 x 400. → All of your images are of same size 512x512.

0000.

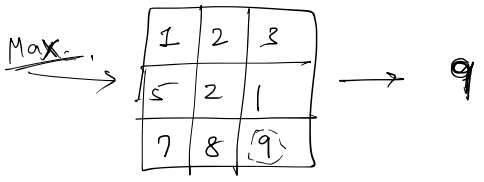
2) Rescaling : 0-255 / 255 → [0-1]

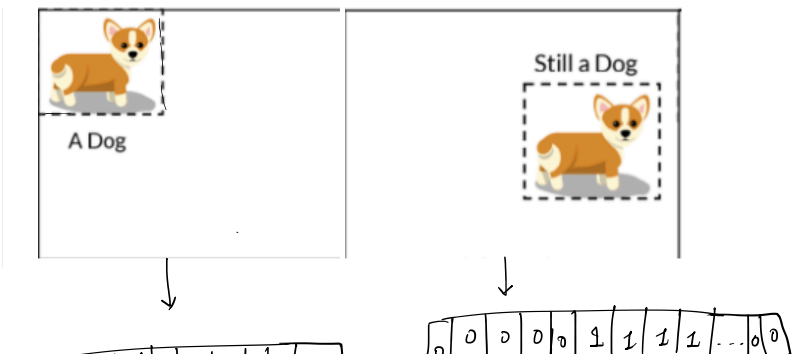
100/255
10/255
255/255
→ 0-1.

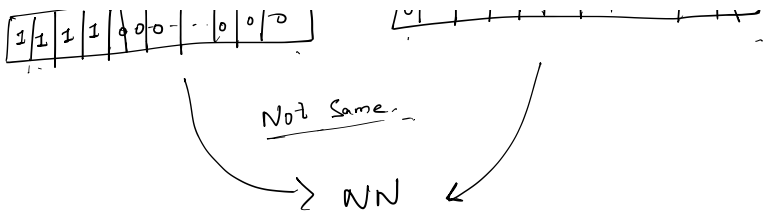


1000 X 49152.

A ___ is a preprocessing step which rescales input values to a new range. → Rescaling

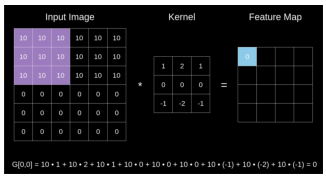






Which of the following statements is False with regards to MLP for image data as input ?

- MLP's usually have lesser parameters when compared to CNN's because they are fully connected → false


Input Image
Kernel
Feature Map
10 10 10 10 10 10
10 10 10 10 10 10
10 10 10 10 10 10
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0
1 2 1
0 0 0
-1 -2 -1
*
=
0
G[0,0] = 10 • 1 + 10 • 2 + 10 • 1 + 10 • 0 + 10 • 0 + 10 • 0 + 10 • (-1) + 10 • (-2) + 10 • (-1) = 0